

Advanced Animal Movement Modelling Training Workshop

Behavioural State Modelling

16 September 2026

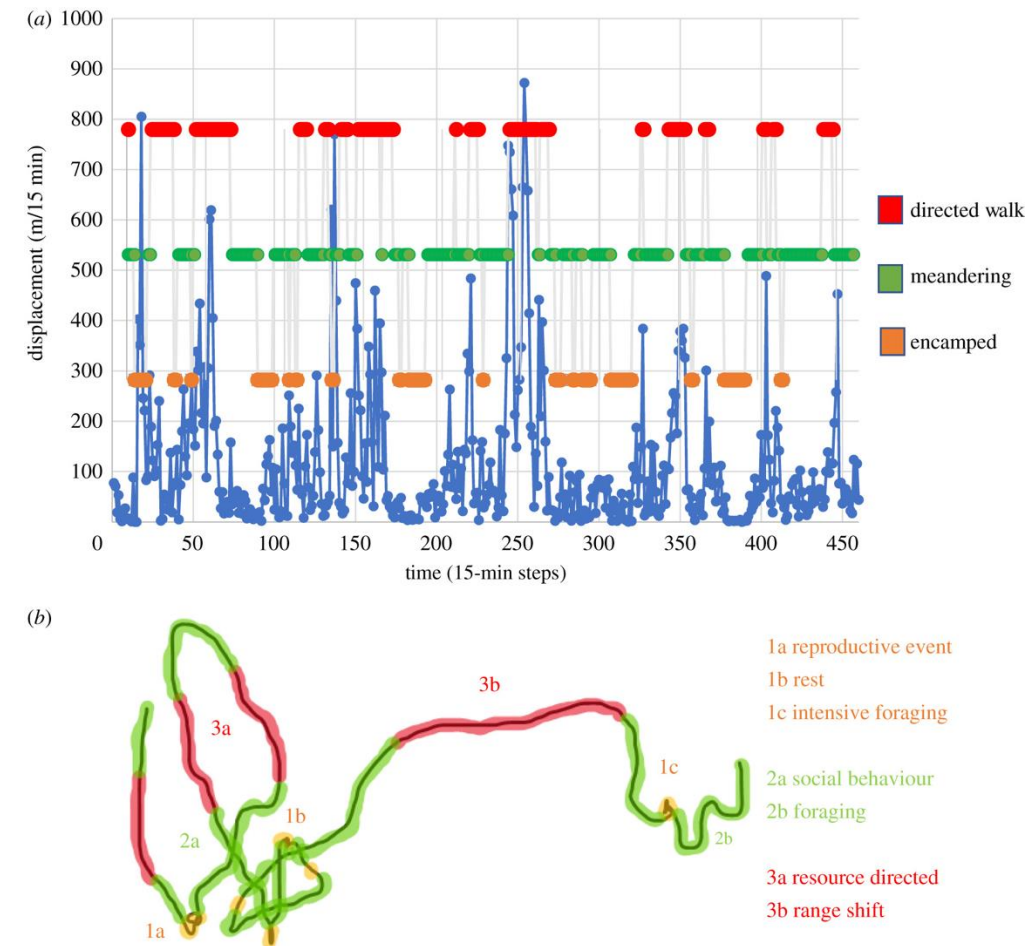
Josh Cullen



Photo credit: Larry Li

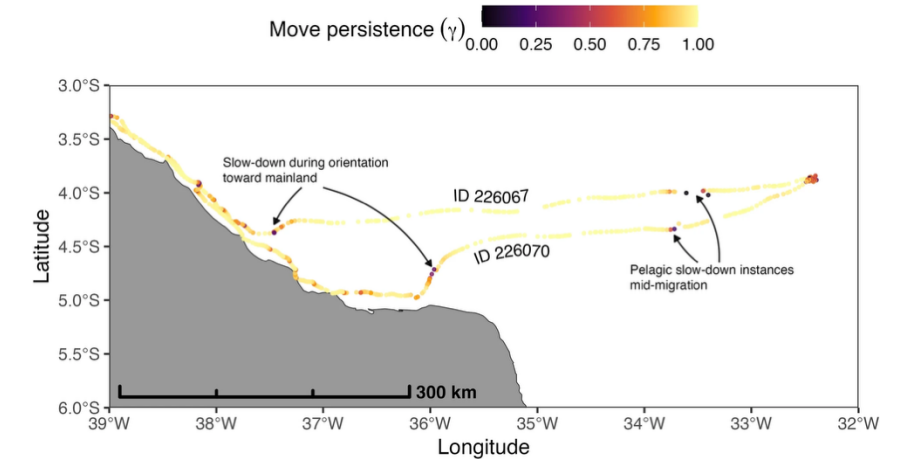
Why the focus on 'states' instead of behaviors?

- When using telemetry and/or biologging data, we often don't know what the animal is actually doing
 - Need to infer the 'behavioral states' from the results, which may represent >1 behavior
 - Helps to have a priori understanding of study species
 - Exception for continuously observed animals (in captivity) or animals that have also been fitted w/ a camera

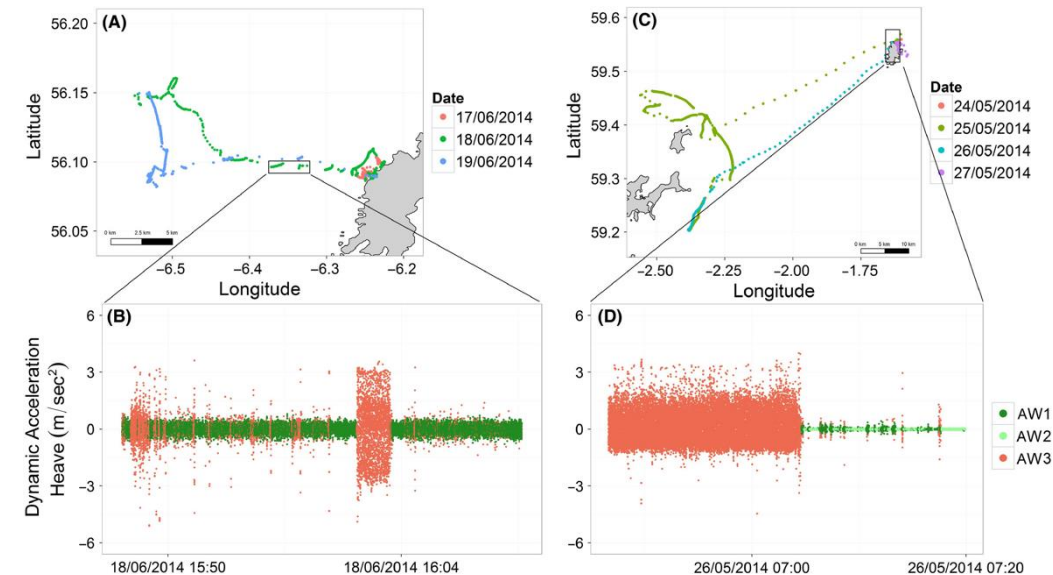


Commonly used methods

- Recently, mechanistic movement models tend to be favored most
 - Which includes state-space models (SSMs) and hidden Markov models (HMMs)
- Mixture models used occasionally
- High temporal resolution data from accelerometers and magnetometers often analyzed with machine learning (ML) methods



Cullen et al. 2025



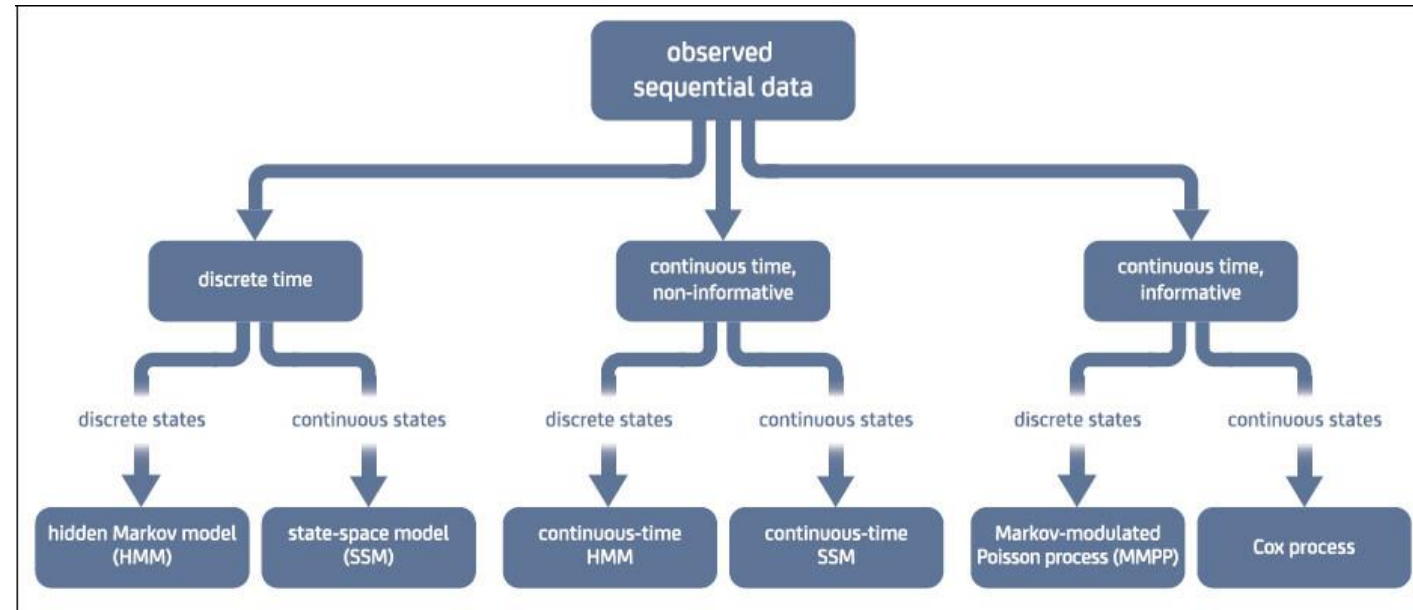
Chimienti et al. 2016

Classes of latent variable models

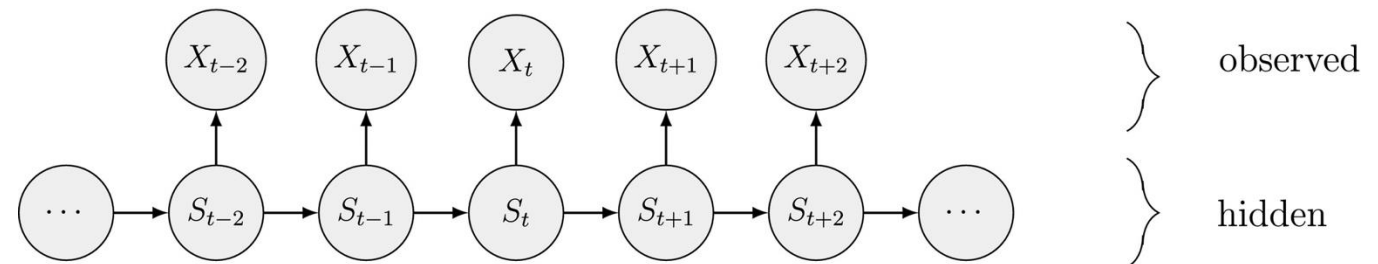
	State space	
	Continuous	Discrete
Temporal dependence	State-space model	Hidden Markov model
Temporal independence	Continuous mixture model	Finite mixture model

Selecting a latent Markov model

- Often uses a first-order Markov assumption
 - I.e., current state (S_t) dependent on previous state (S_{t-1})
 - Typical of modern telemetry data
- If assumption violated, could possibly apply higher-order Markov model or semi-Markov model



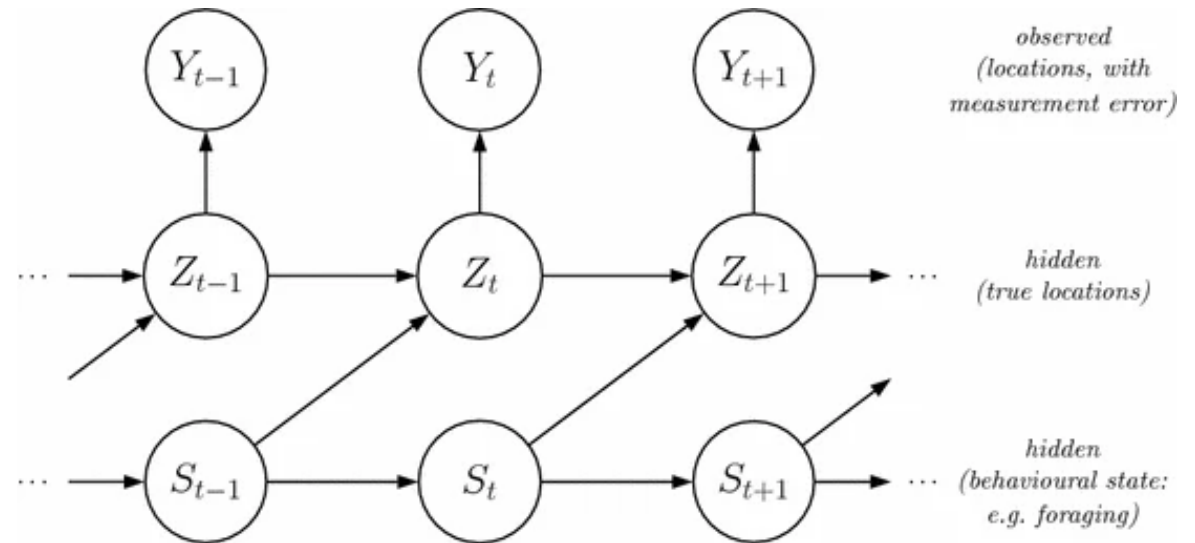
Mews et al., 2025



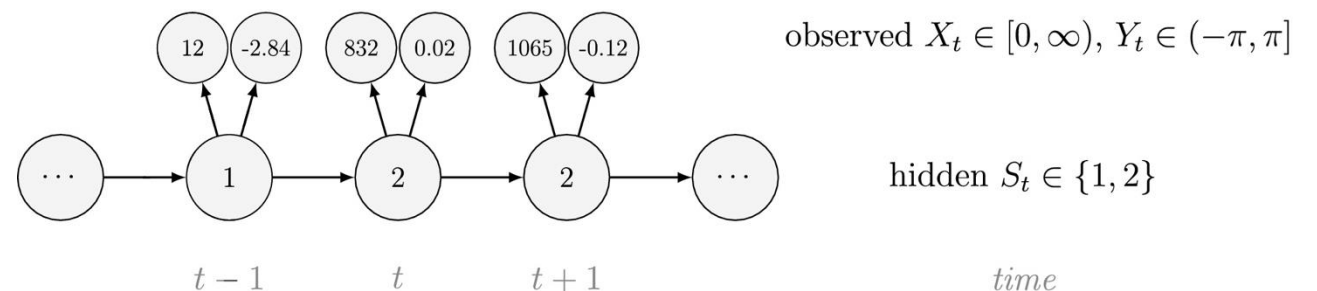
McClintock et al., 2020

Selecting a latent Markov model

- SSMs can estimate likely locations w/ measurement error and simultaneously estimate behavioral states
 - Estimates continuous state-space (behavior on continuum)
- HMMs typically assume negligible measurement error and rely on regular sampling intervals
 - Although can account for violations via other means
 - Estimate states in finite space

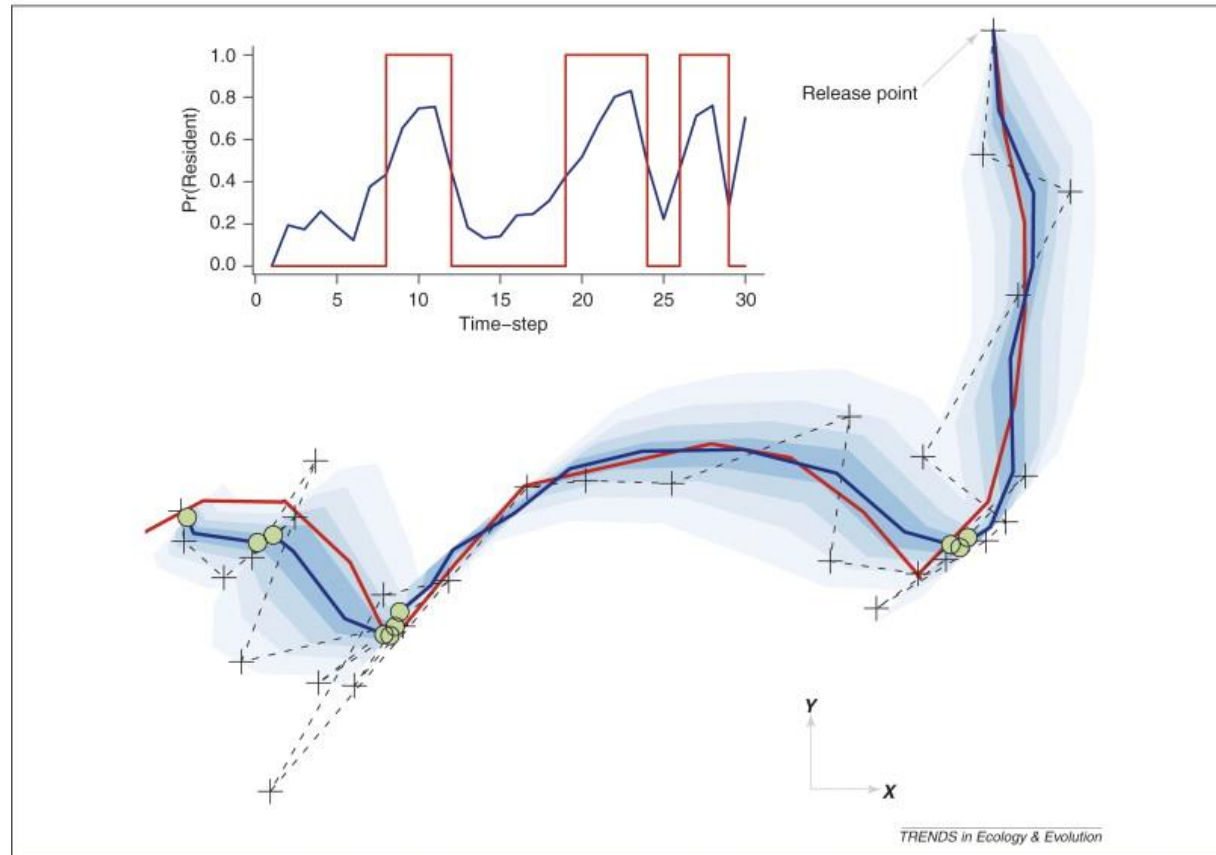


Patterson et al., 2017



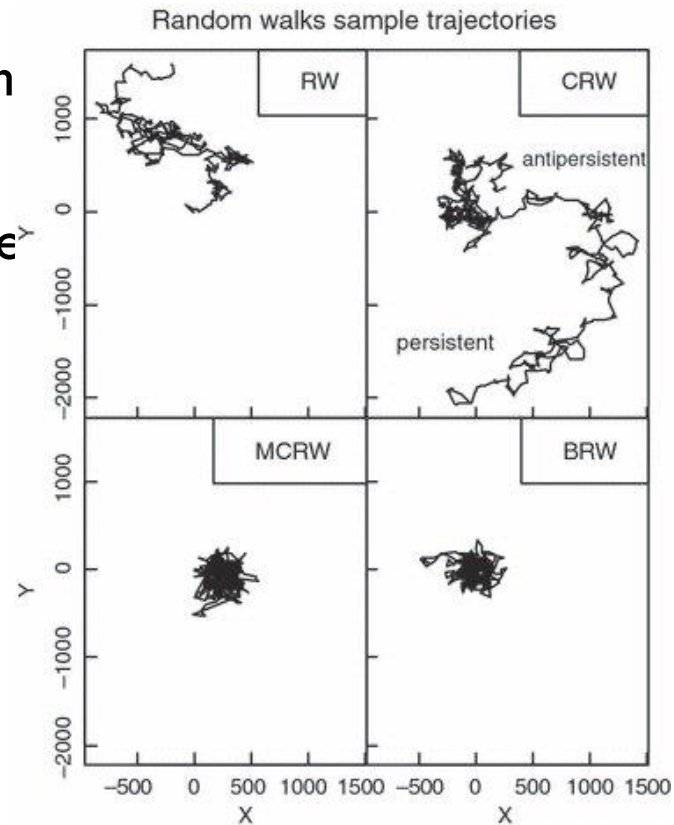
McClintock et al., 2020

Example of SSM in action

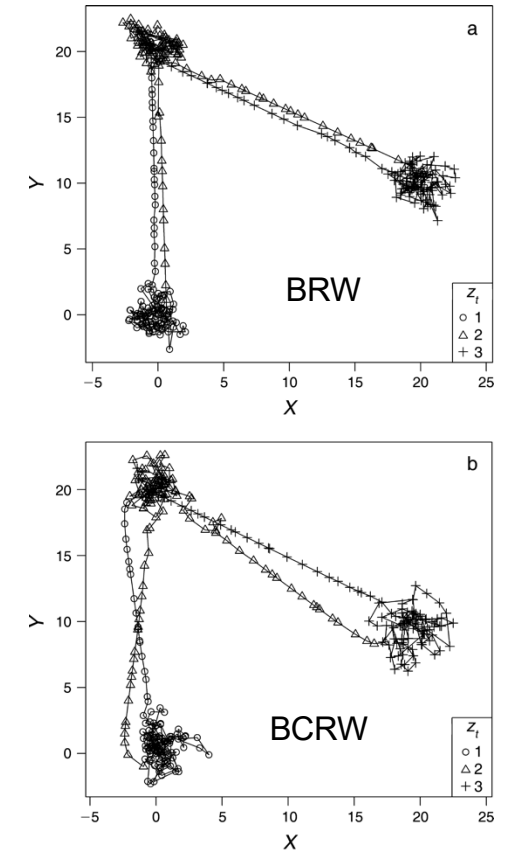


Selecting a latent Markov model

- SSMs and HMMs often assume movement follows a correlated random walk (CRW)
 - Other movement models available
- Some of the most common alternatives include:
 - Random walk (RW)
 - Biased RW (BRW)
 - Biased CRW (BCRW)



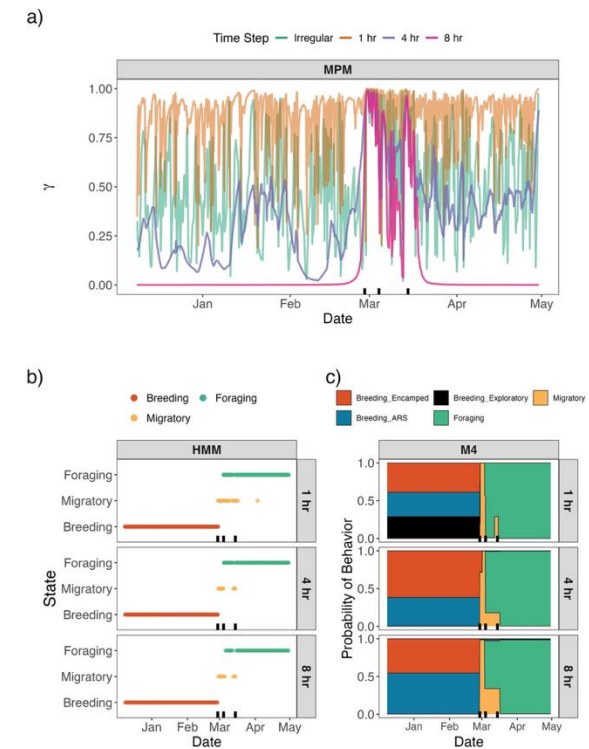
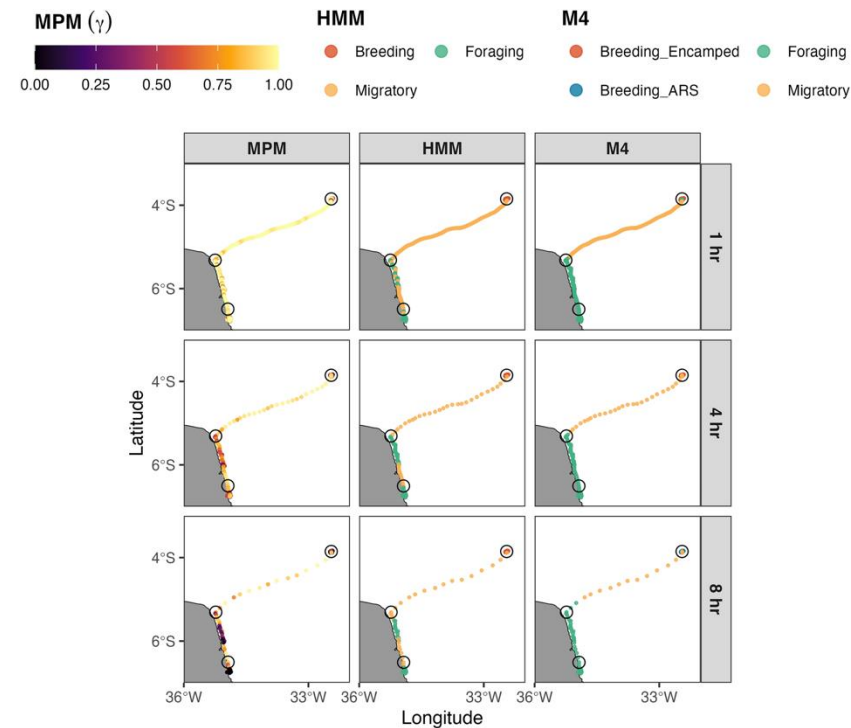
Börger et al. 2008



McClintock et al., 2012

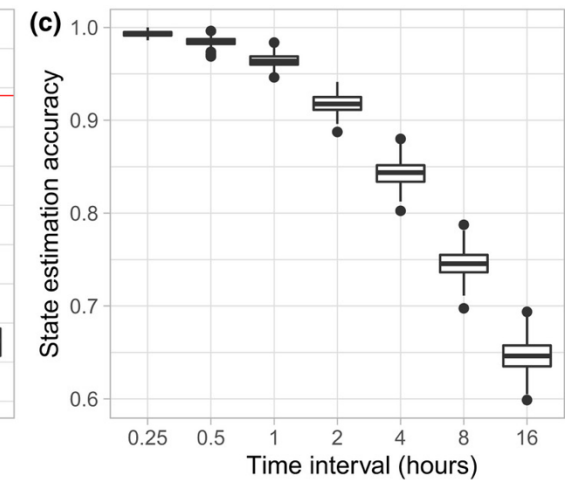
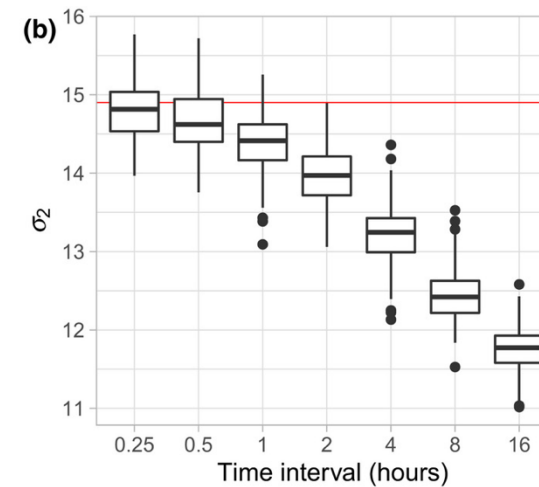
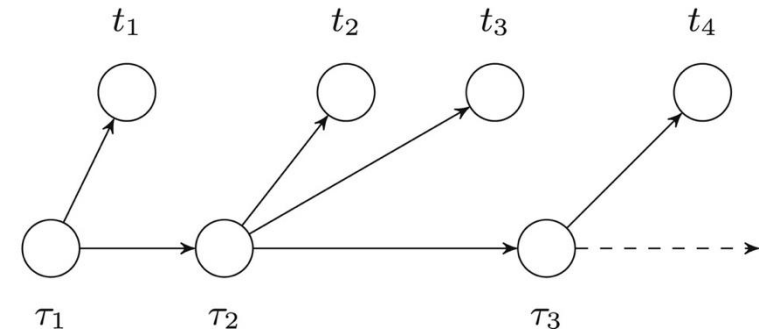
Importance of temporal resolution

- Related to scale of state inference and the state dwell time (i.e., how long an animal spends in a given state)
- Coarser time intervals will estimate coarser states
 - The opposite is also generally true
- Important consideration before tags are ever deployed



Importance of temporal resolution

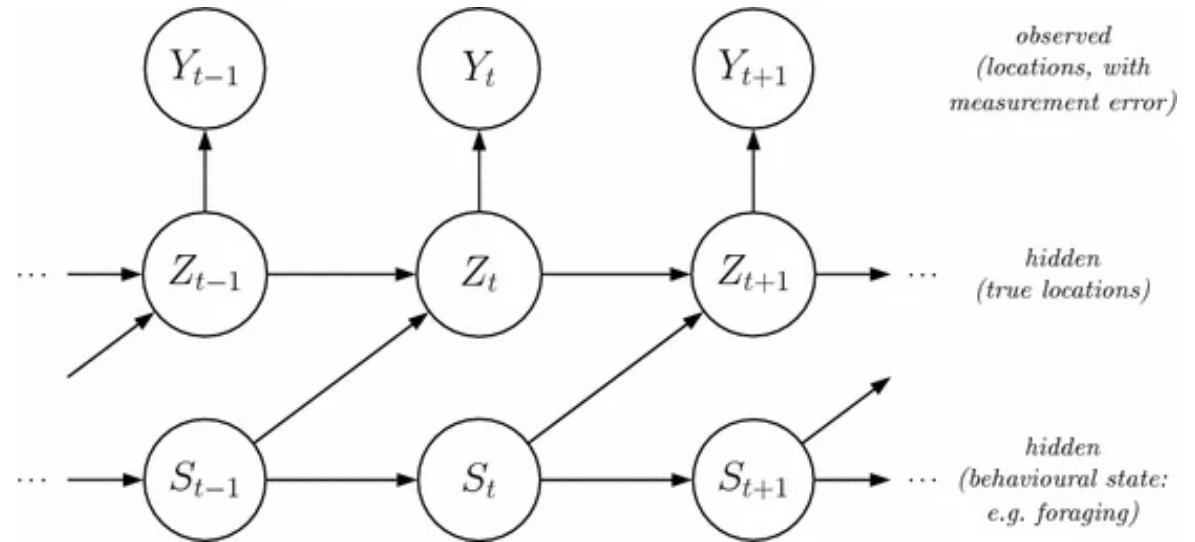
- Continuous-time HMMs assume “snapshot property”
 - value of the observation process at time t only depends on the state at that time (Patterson et al., 2017; Glennie et al., 2023)
- This may be reasonable approx. at high sampling intervals relative to state switches
- **Rule of thumb:** sampling interval should be shorter than shortest expected state dwell time



State-space models

Breaking down a telemetry SSM

- Consists of observation and process models
 - Observation: locations of animal movement
 - Process: “true” locations and behavioral states
- Most frequently applied on Argos and light-level geolocation data w/ large location errors
- Available R packages (`bsam`, `aniMotum`) provide discrete- and continuous-time models, respectively



Patterson et al., 2017

$$\mathbf{z}_t = \mathbf{z}_{t-1} + \gamma_{b_{t-1}} \mathbf{T}(\theta_{b_{t-1}})(\mathbf{z}_{t-1} - \mathbf{z}_{t-2}) + \epsilon_t$$

$$\mathbf{y}_i = (1 - j_i)\mathbf{z}_{t-1} + j_i\mathbf{z}_t + \eta_i$$

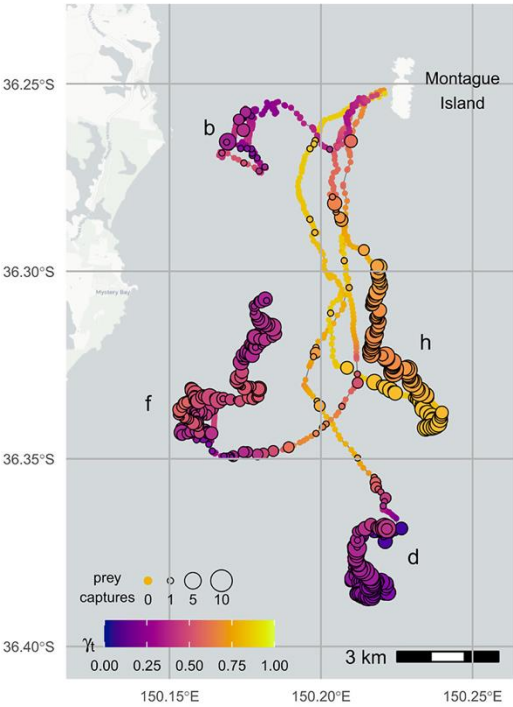
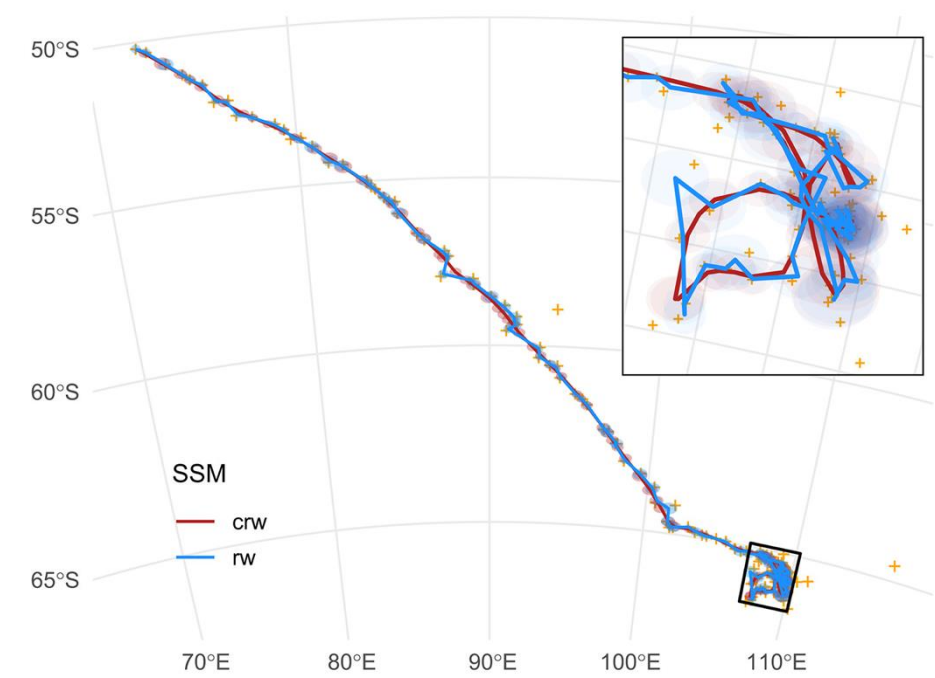
Methods to fit SSM

TABLE 2. Comparison of the fitting methods discussed in *Fitting SSMs*.

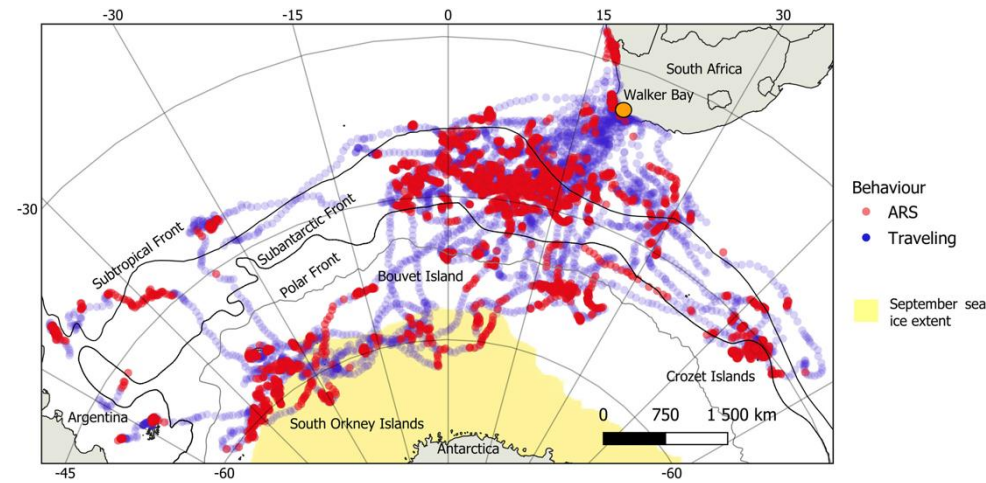
Method	Framework	Pros	Cons	R package
Kalman filter and MLE	Frequentist	Efficient and exact	Only applicable to linear Gaussian SSMs	dlm, MARSS
Laplace approximation	Frequentist	Efficient and flexible	States need to be approximable with a continuous unimodal distribution (e.g., no discrete states)	TMB ← aniMotum
Particle filter and iterated filtering	Frequentist	Flexible	Can be slow and sensitive to starting values	pomp, nimble
MCMC-MH	Bayesian	Flexible	Can be slow and sensitive to convergence problems	rjags, nimble, R2WinBUGS, BRugs ← bsam
MCMC-HMC	Bayesian	Efficient and flexible	Require continuous parameters and states or marginalization	rstan
Information reduction	Bayesian	Flexible and fewer estimation problems	Can be slow and imprecise	EasyABC

Notes: HMC, Hamiltonian Monte Carlo; MCMC, Markov chain Monte Carlo; MH, Metropolis–Hastings; MLE, maximum-likelihood estimate.

Common examples of SSM use



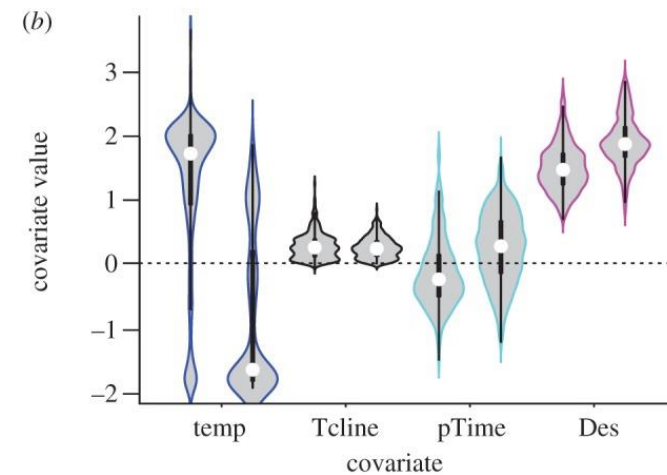
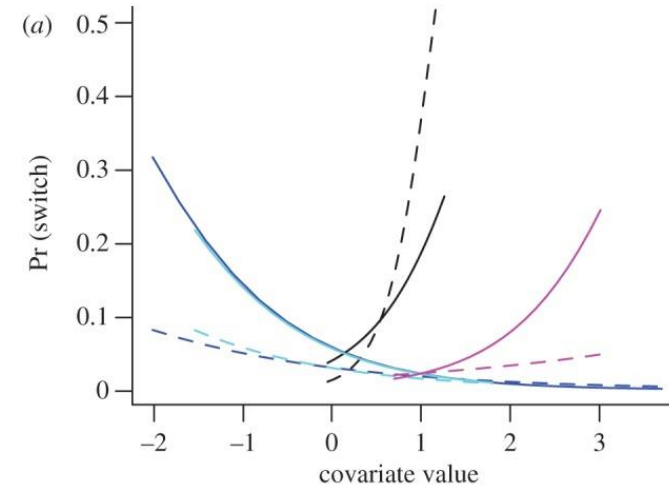
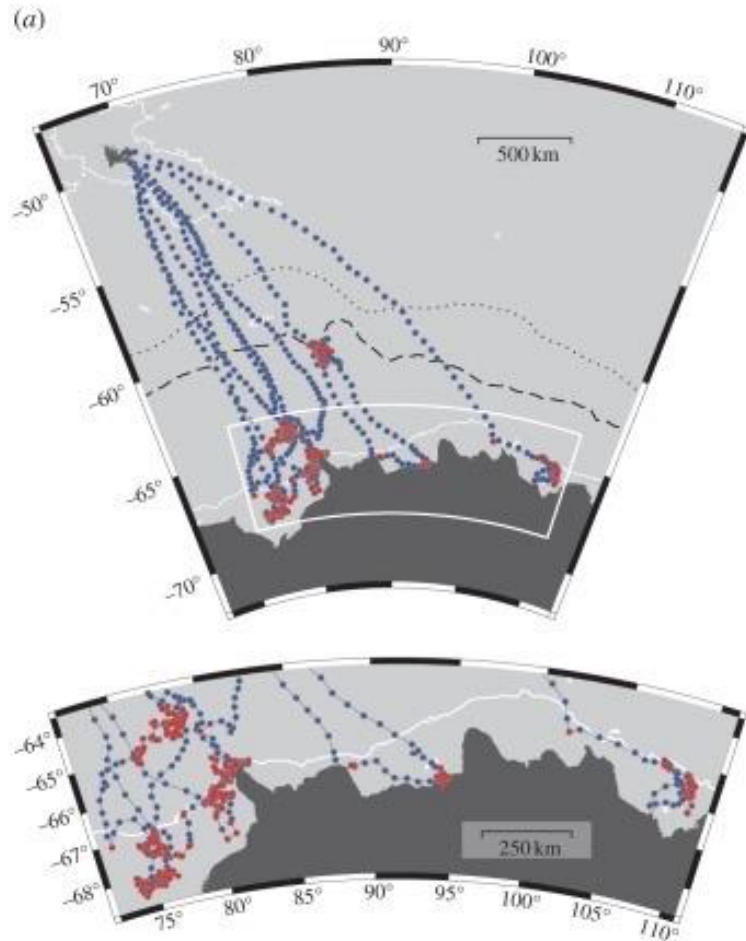
Jonsen et al., 2023



Germishuizen et al., 2026

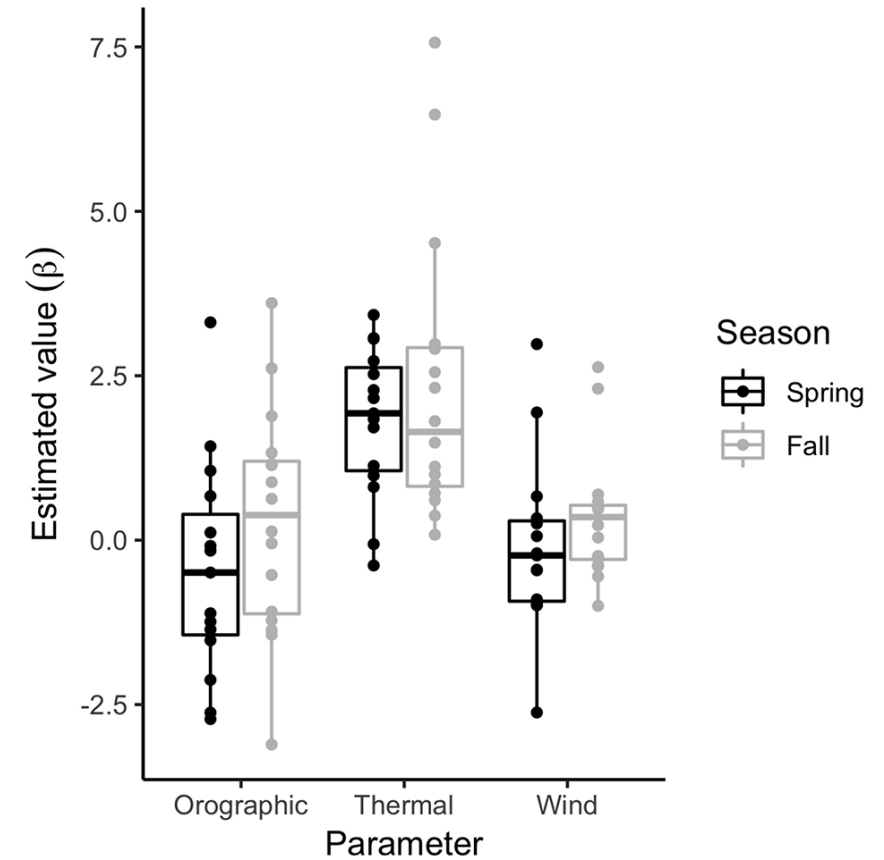
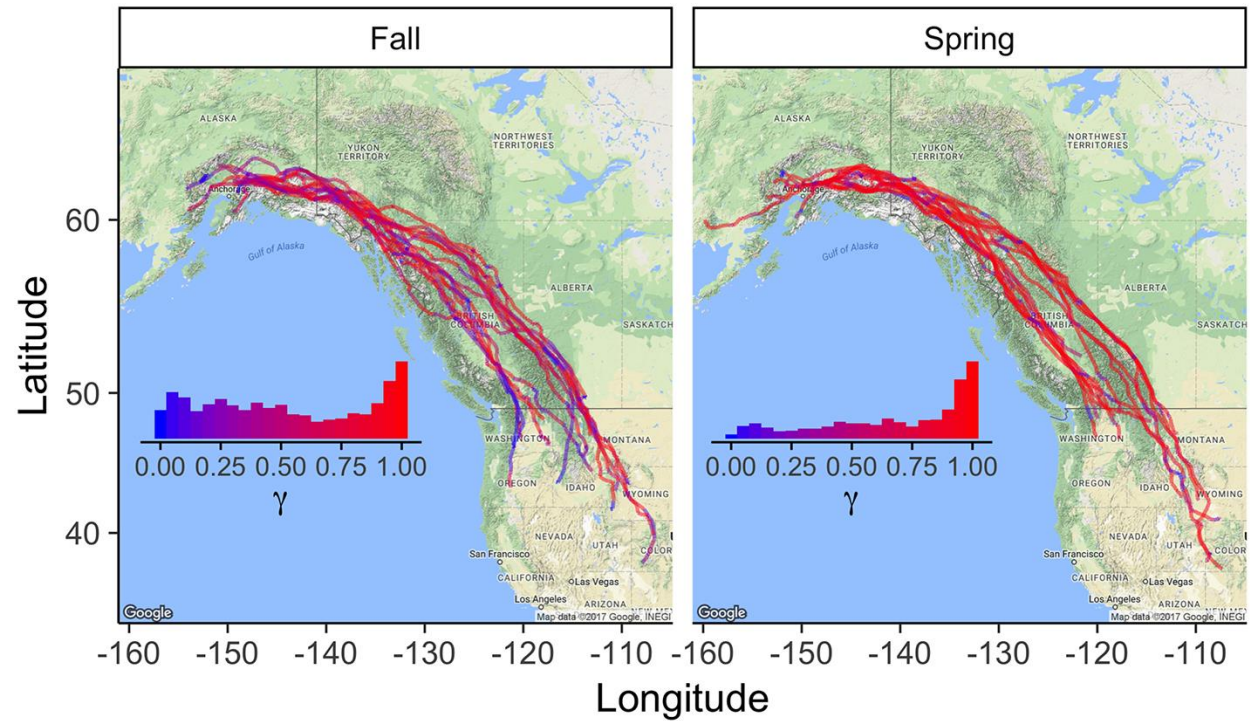
Inclusion of environmental covariates

Example 1: southern elephant seals (Bestley et al. 2013)



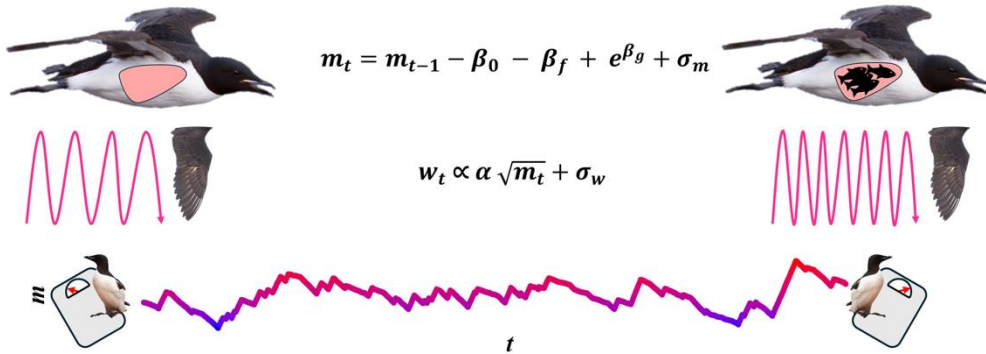
Inclusion of environmental covariates

Example 2: golden eagles (Eisaguirre et al. 2019)

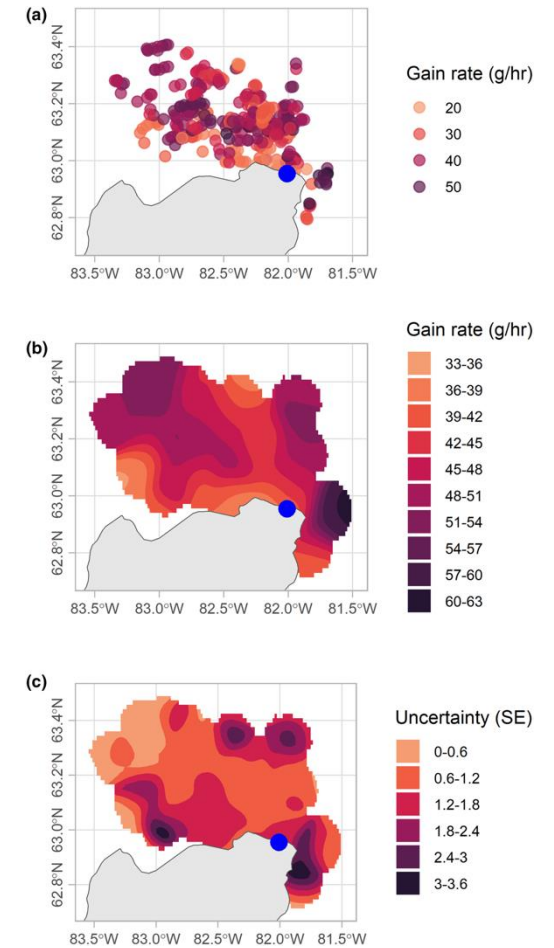
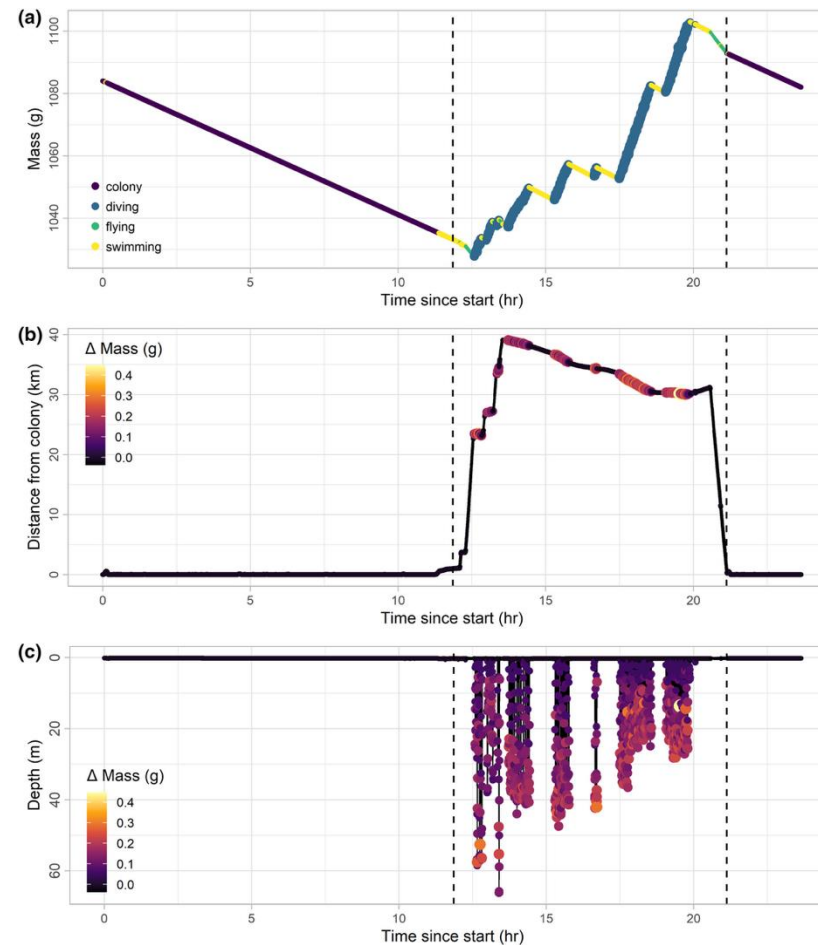


Estimating physiological changes

Example 1: thick-billed murres (Patterson et al. 2026)

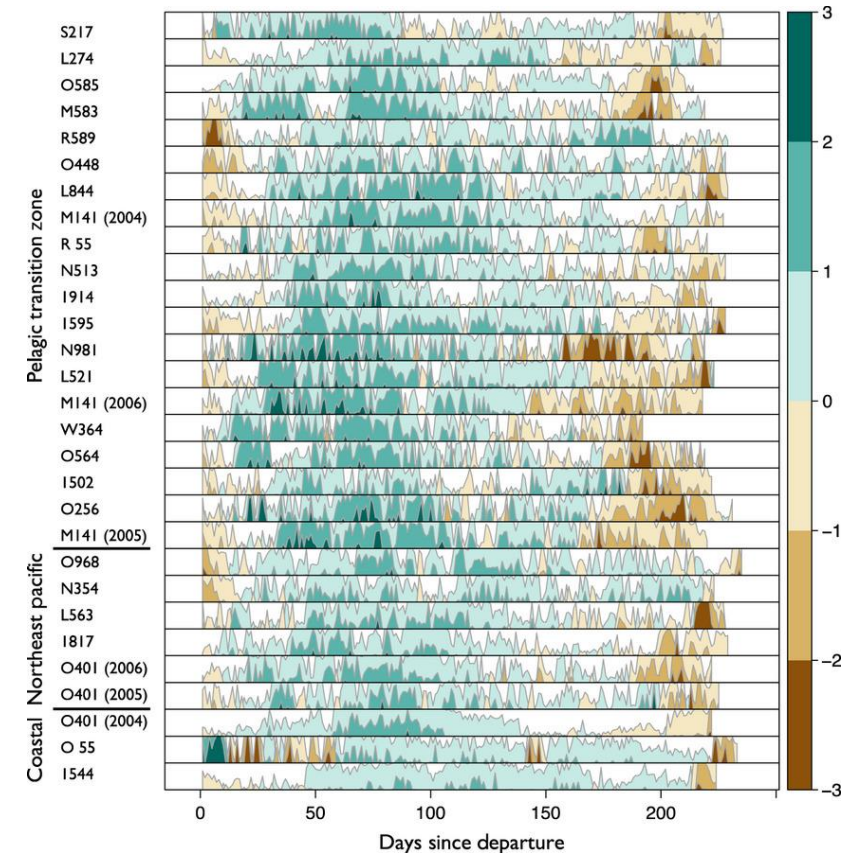
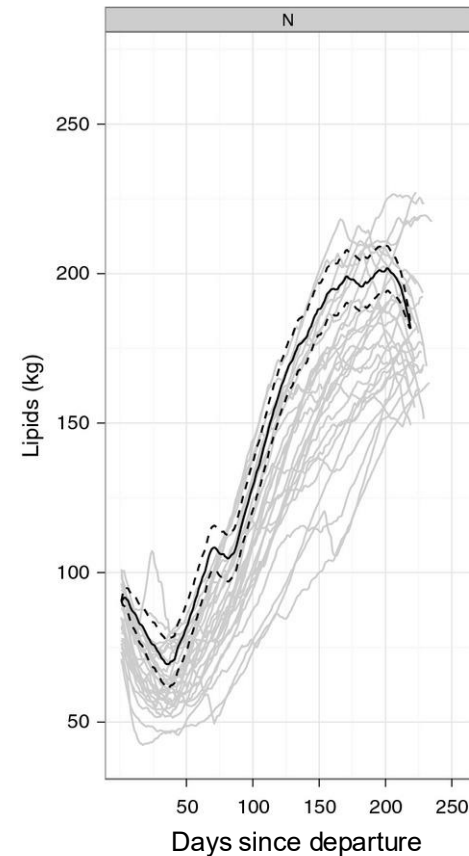
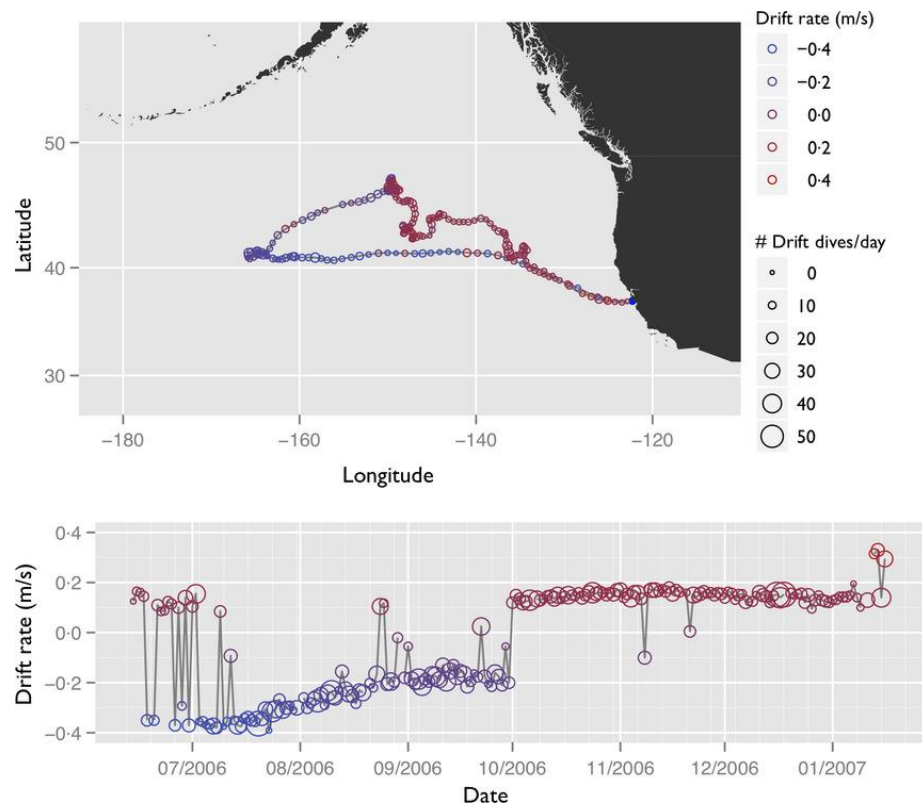


Wing-beat frequency --> body mass changes



Estimating physiological changes

Example 2: northern elephant seals (Schick et al. 2013)

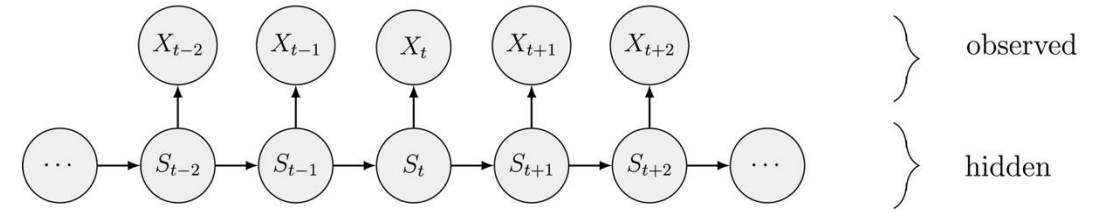


Dive drift rate --> body mass and lipid changes

Hidden Markov models

Breaking down a telemetry HMM

- Also consists of observation and process models
 - Observation: movement metrics (e.g., step lengths, turning angles)
 - Process: behavioral states
- Most frequently applied on GPS data w/ negligible location error
- Available R packages (`moveHMM`, `momentuHMM`, `hmmTMB`) provide discrete-time models
 - But dev version of `momentuHMM` also has a continuous-time HMM available
- General advice: # of data streams + 1 likely reflects max states that can be estimated



McClintock et al., 2020

Initial state distrib.

$$\delta = (P(S_1 = 1), \dots, P(S_1 = N))$$

Transition probability matrix (t.p.m.)

$$\Gamma = \begin{pmatrix} \gamma_{11} & \dots & \gamma_{1N} \\ \vdots & \ddots & \vdots \\ \gamma_{N1} & \dots & \gamma_{NN} \end{pmatrix}$$

Matrix of state-dep. distribs

$$\mathbf{P}(X_t = x_t) = \begin{pmatrix} f(X_t = x_t | S_t = 1) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & f(X_t = x_t | S_t = N) \end{pmatrix}$$

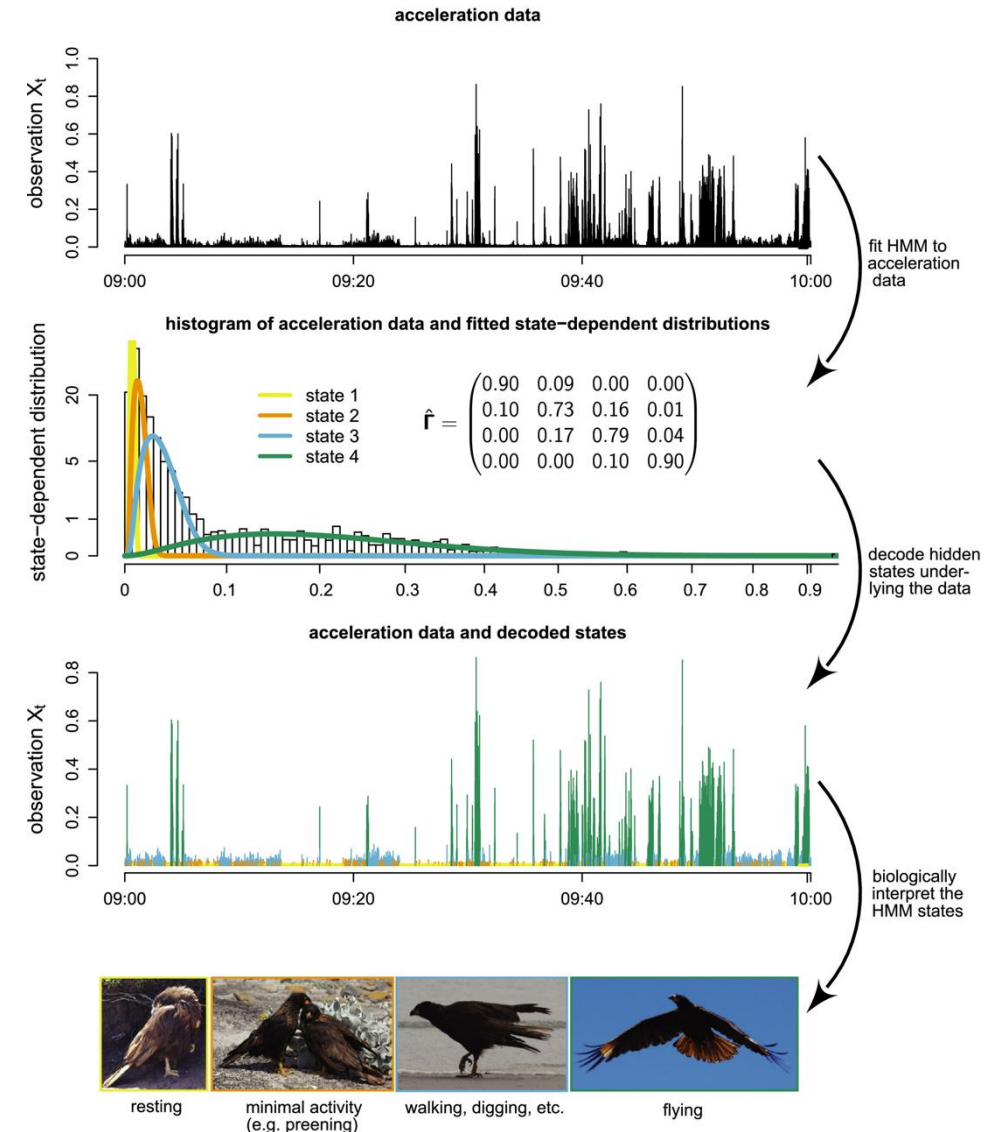
HMM likelihood fun.

$$\mathcal{L} = P(X_t = x_t | \theta) = \delta \Gamma \mathbf{P}(x_1) \dots \Gamma \mathbf{P}(x_T) \mathbf{1}$$

Akeresola et al., 2024

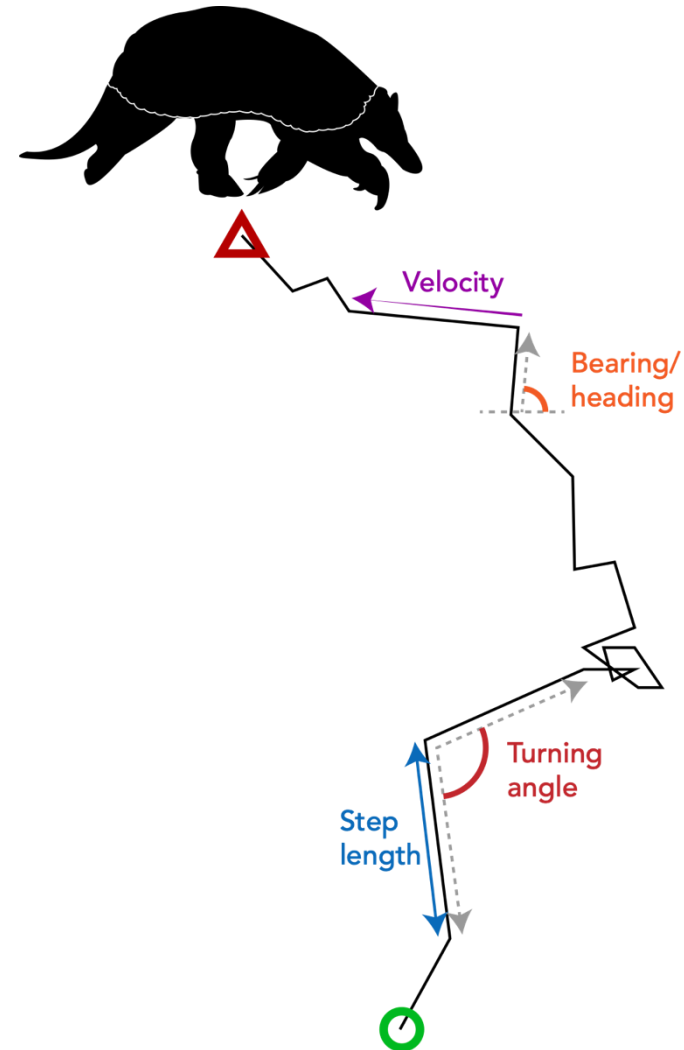
HMMs in practice

- While software for fitting ready-to-go SSMs require minimal up-front decision making, HMMs require more thought
 - Need to select data streams to estimate states
 - Need to decide on probability distributions representing each data stream
 - Need to decide on the number of states to estimate
 - Need to define initial parameter values
- And once model is fitted, need to interpret what the states represent!

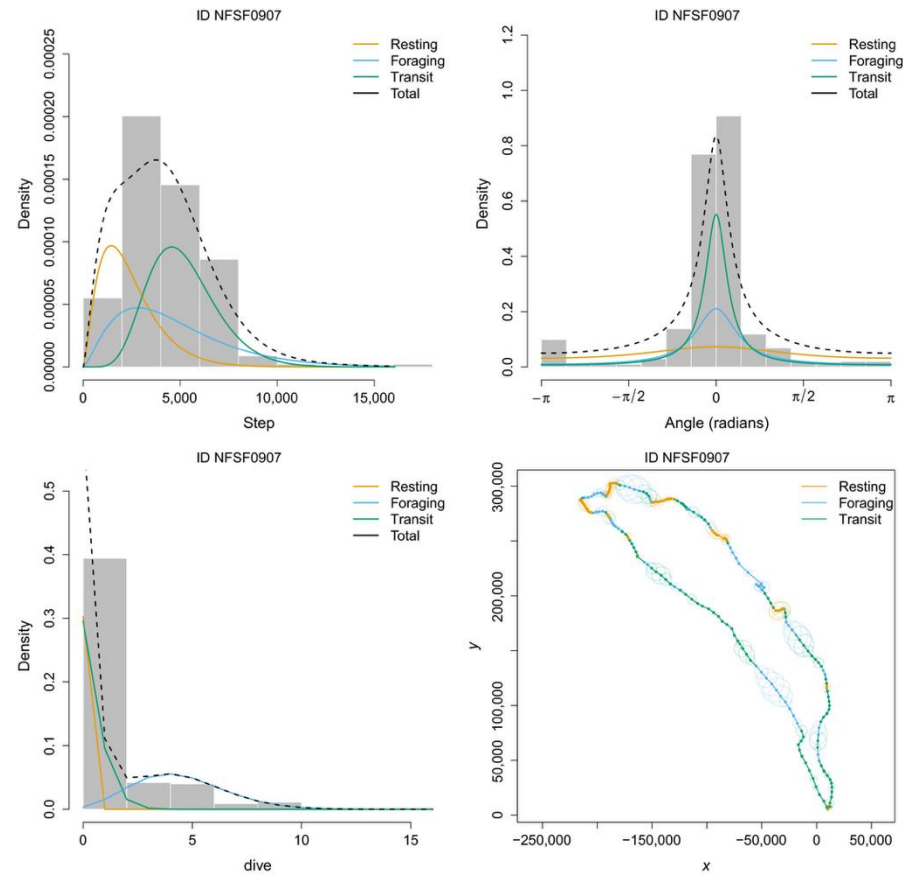


Movement metrics analyzed by HMMs

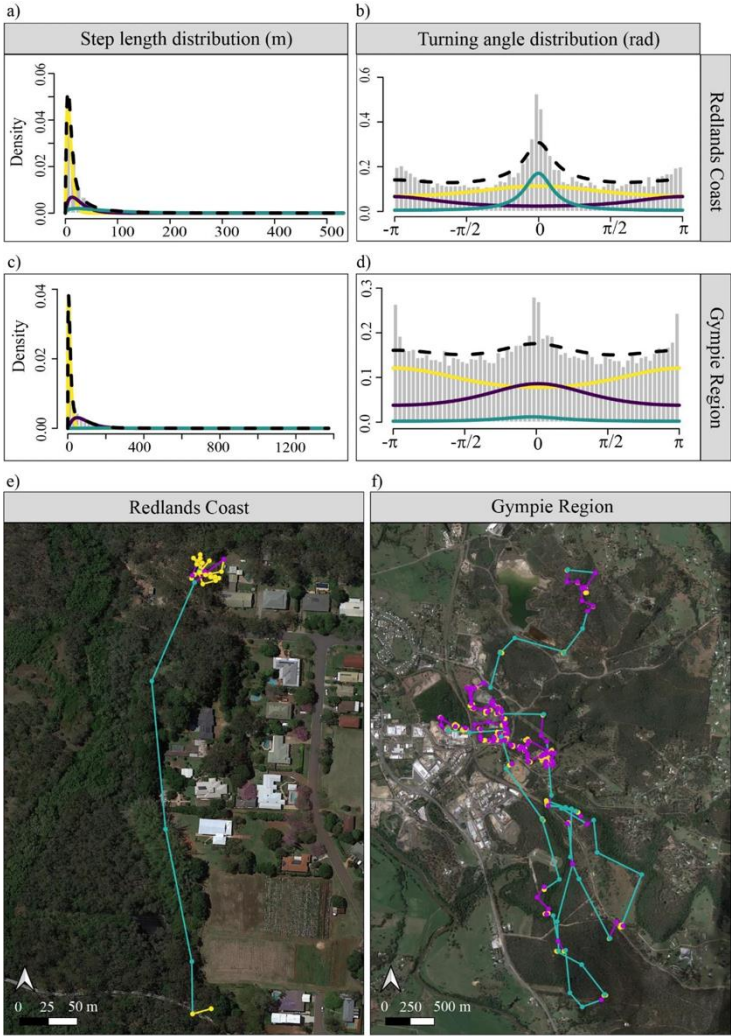
- Bivariate positions or their increments
- Distance between successive observations
- Compass direction (i.e., heading/bearing)
- Changes in direction between successive relocations



Common examples of HMM use



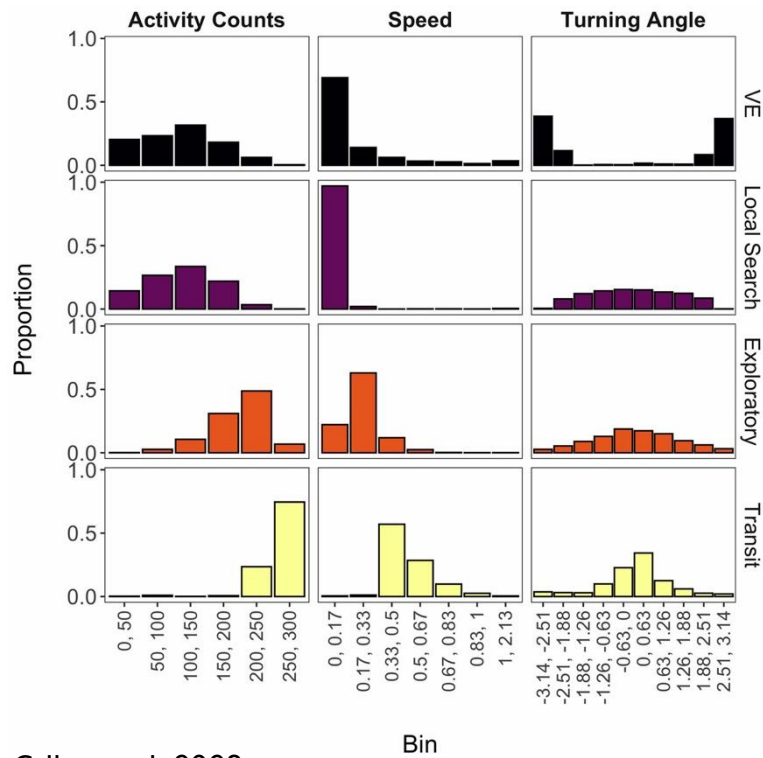
McClintock & Michelot, 2018



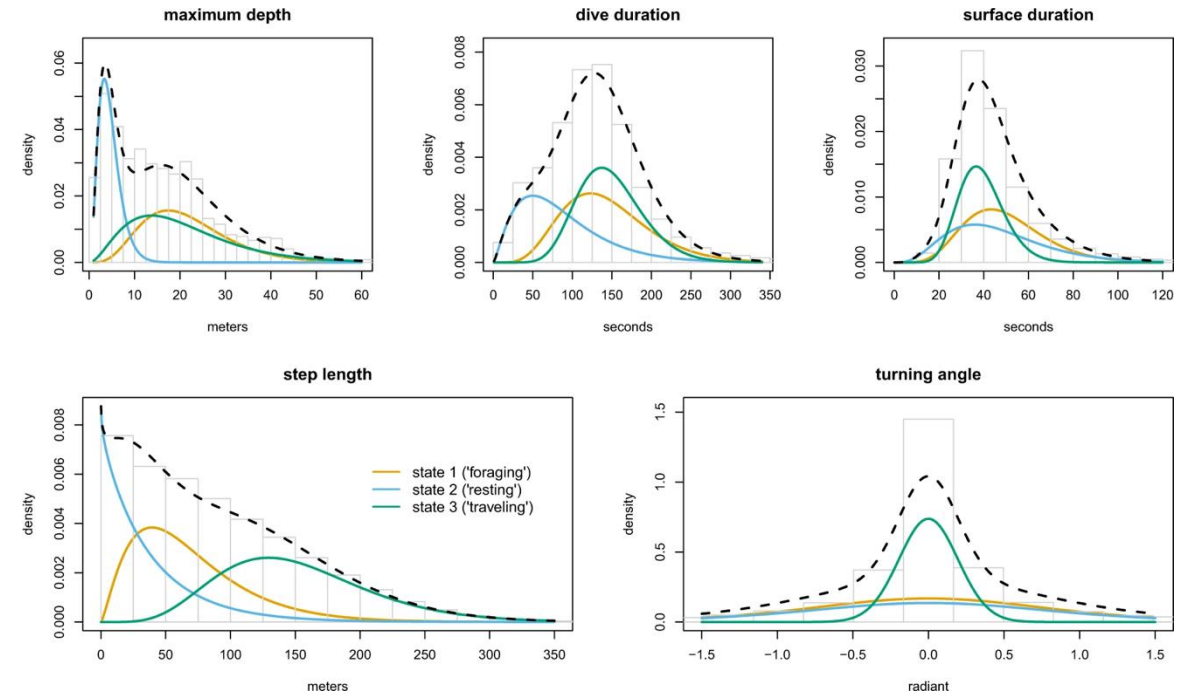
Movement behaviour Slow and tortuous Medium and tortuous Fast and directed

Going beyond step lengths and turning angles

- Other data streams from on-board sensors can also be used to estimate states



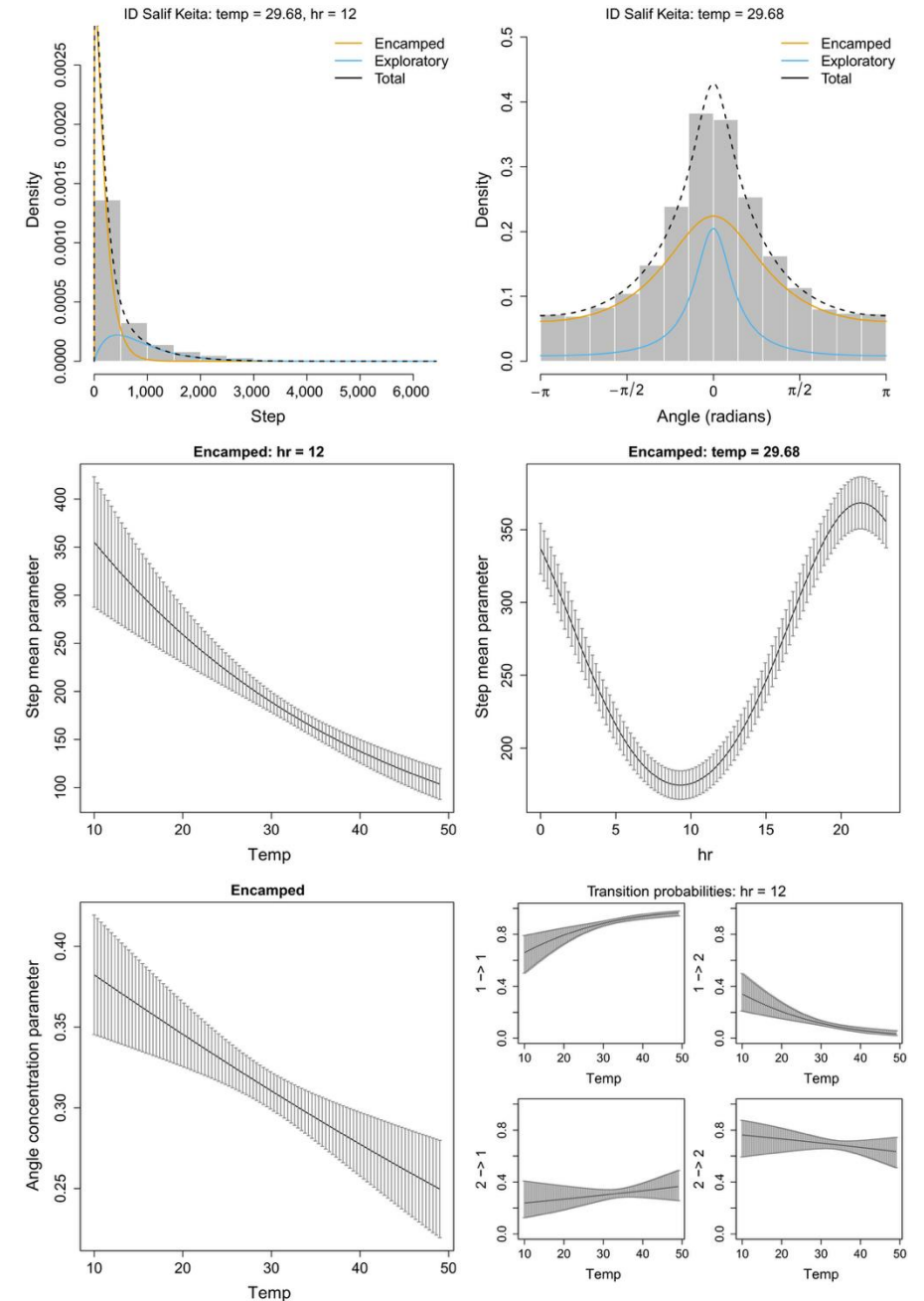
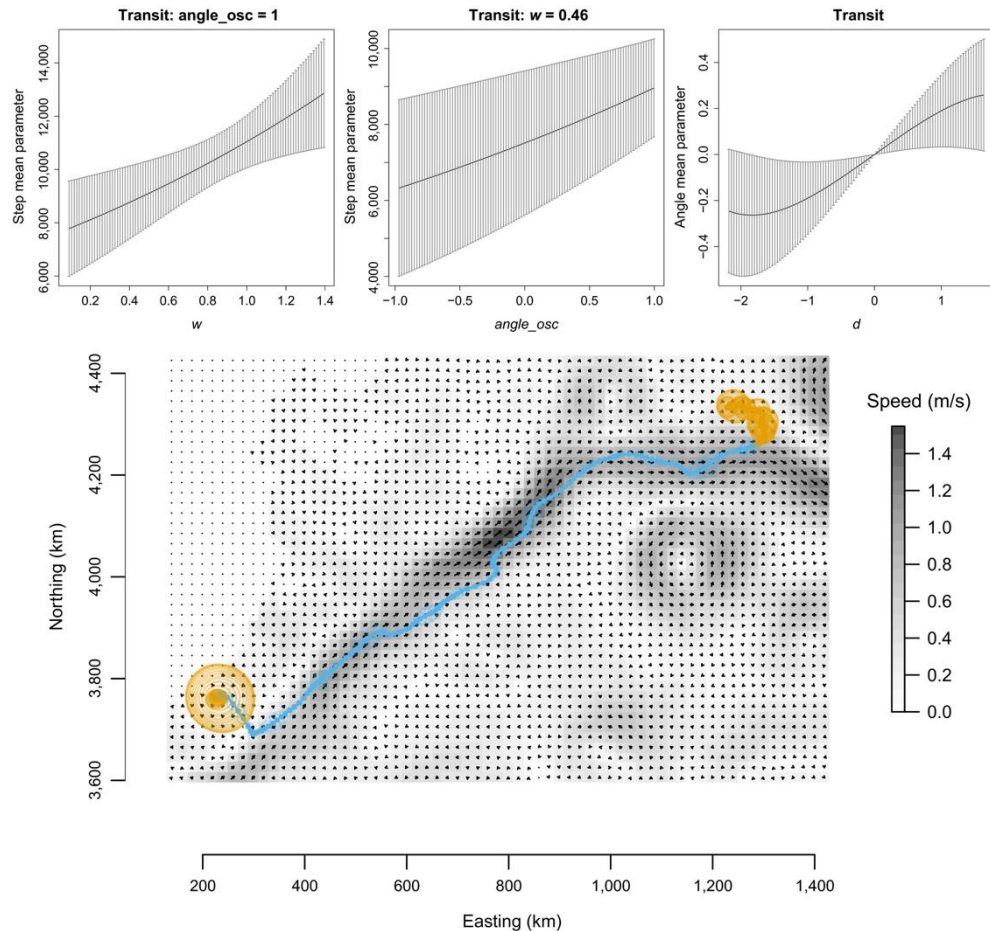
Cullen et al. 2023



van Beest et al. 2019

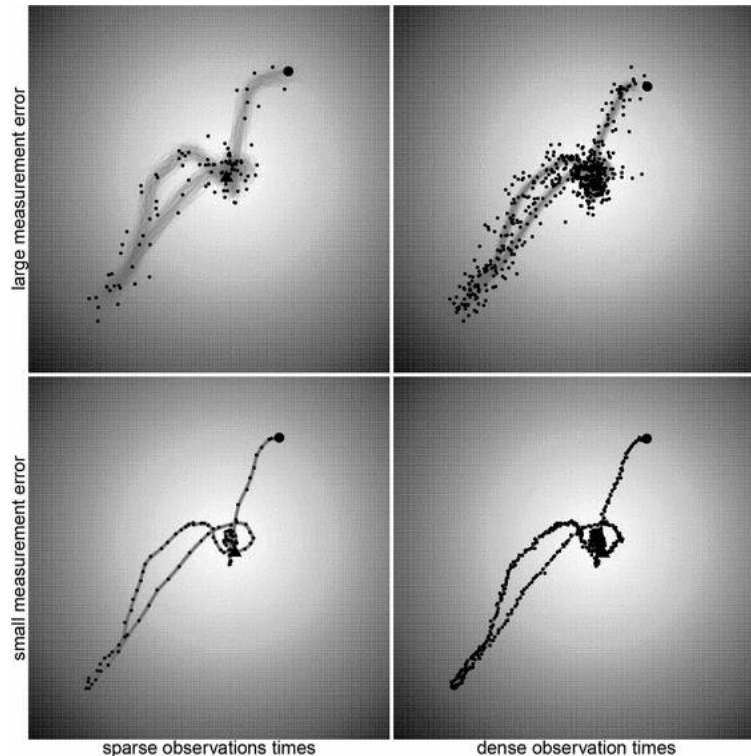
Including covariates

- Can be added to data stream parameters or t.pm.

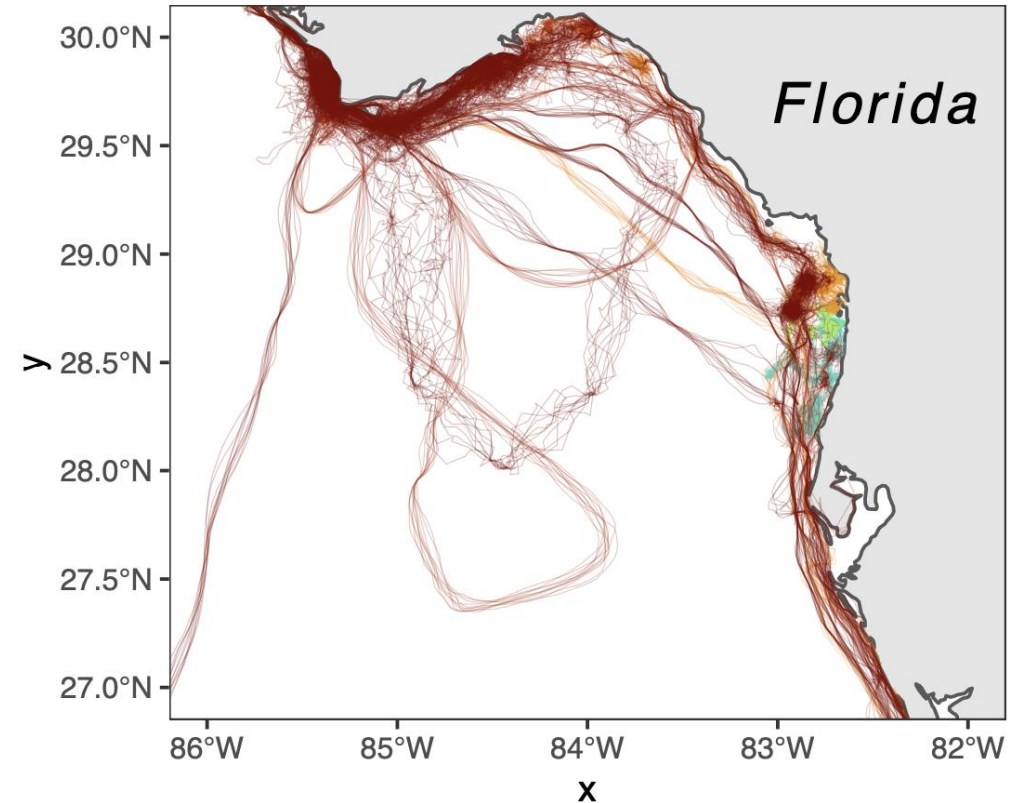


Accounting for irregular time series and/or location error

- Using multiple imputation to account for uncertainty in movement process (McClintock, 2017)

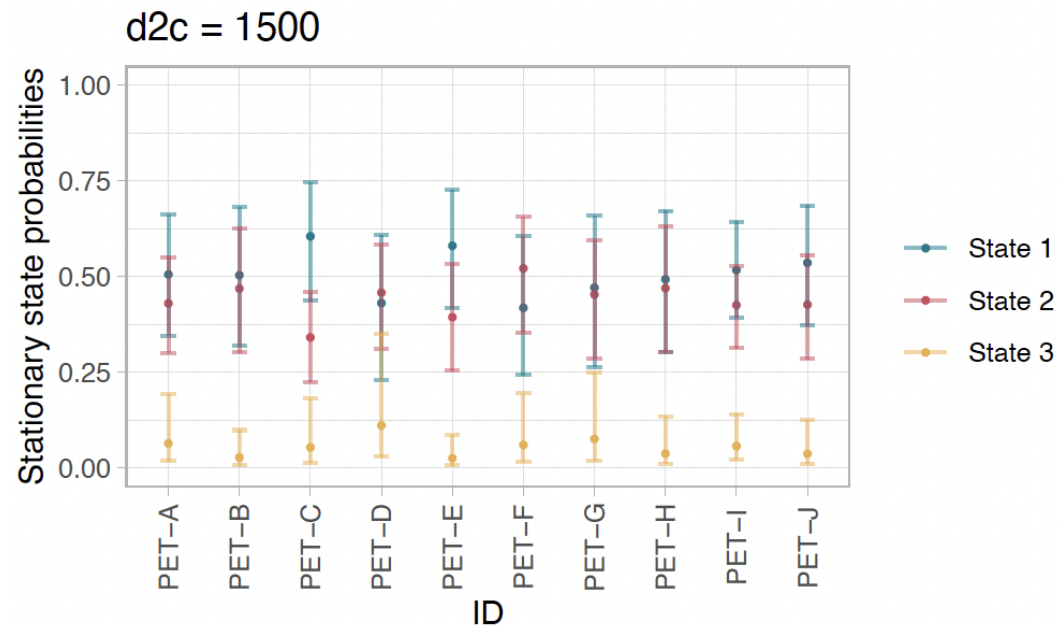


Scharf et al. 2017



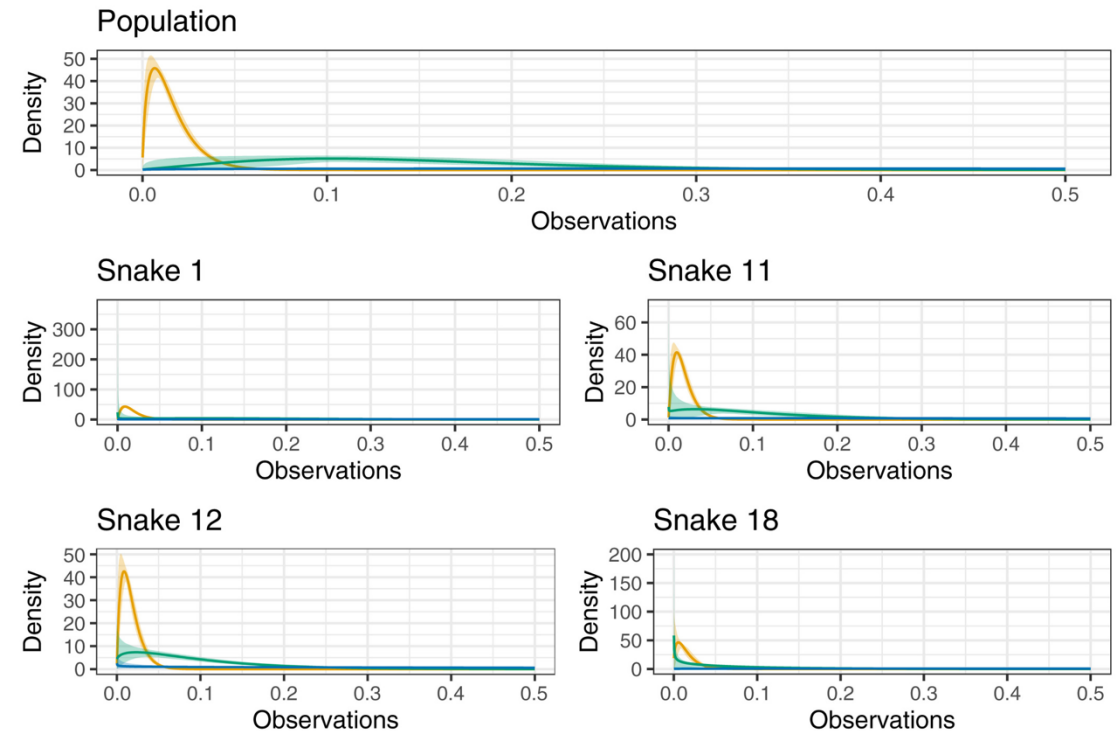
Including random effects

- Often to account for ID or group-level differences in movement
- Generally not very helpful and more difficult to fit (McClintock, 2021)



Michelot 2025

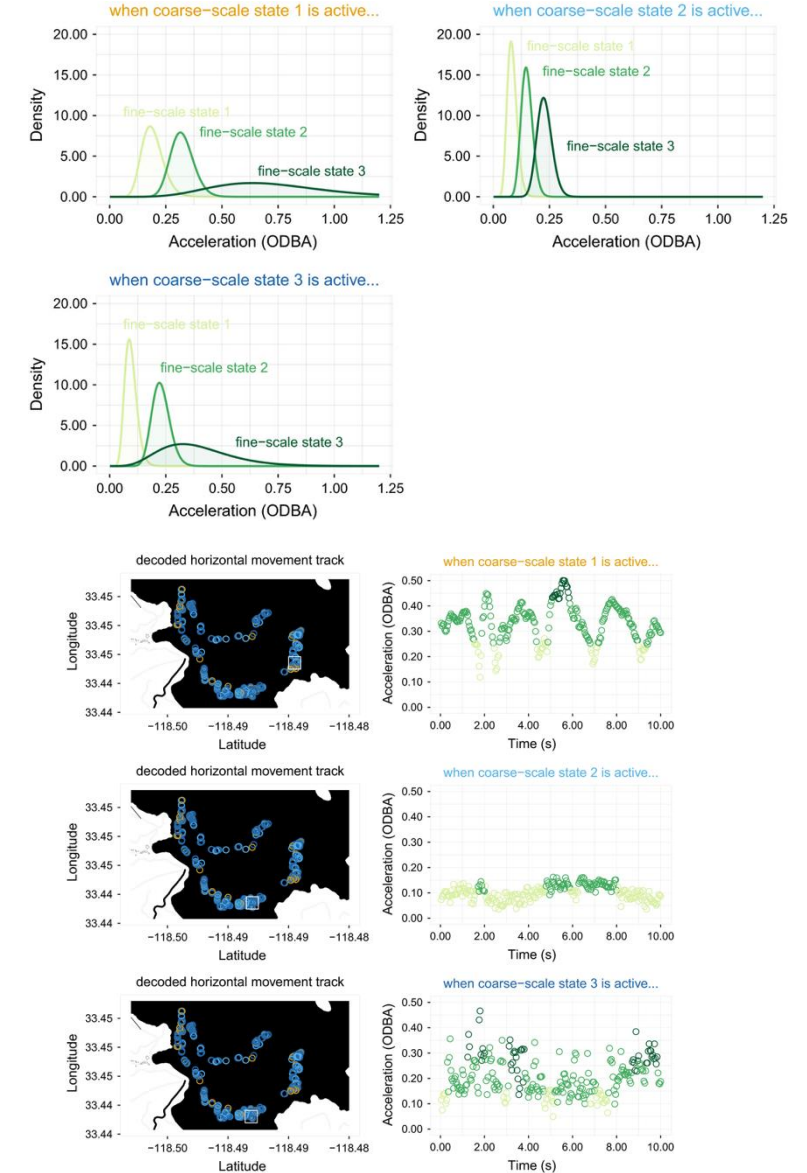
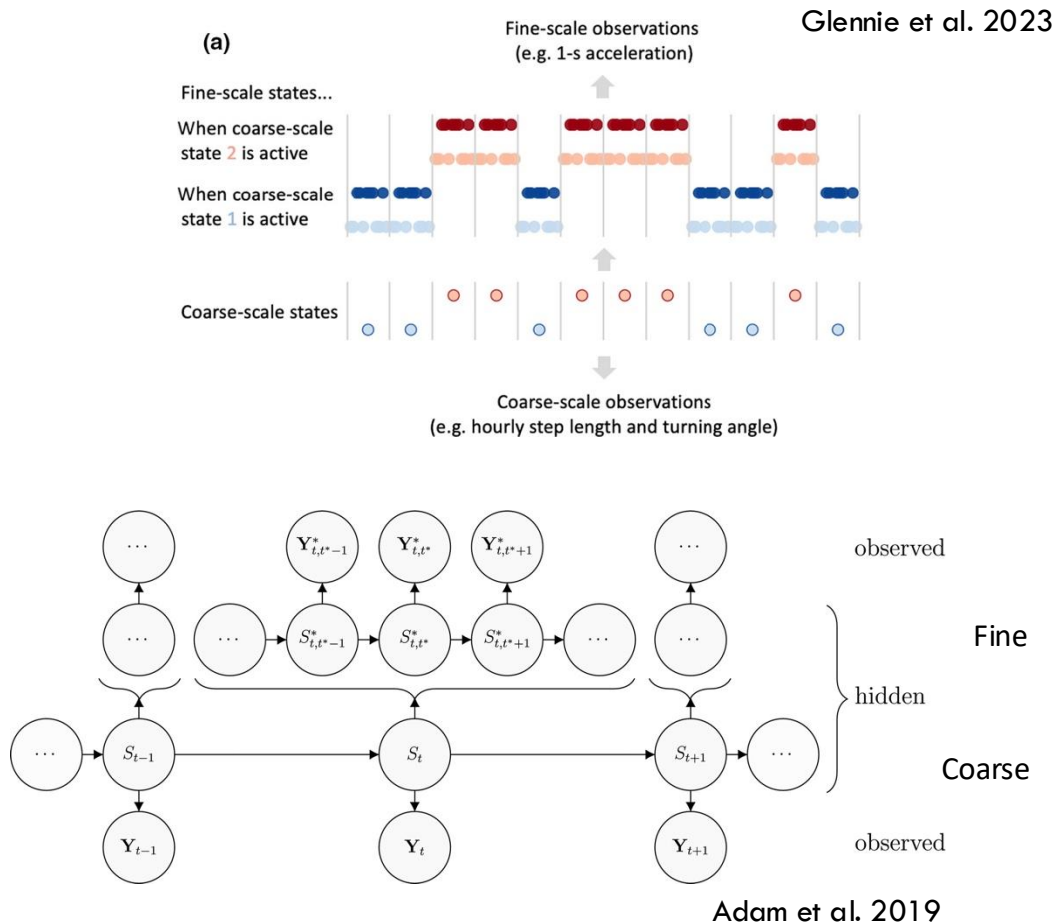
State-dependent distributions (partial pooling)



Glennie et al. 2023

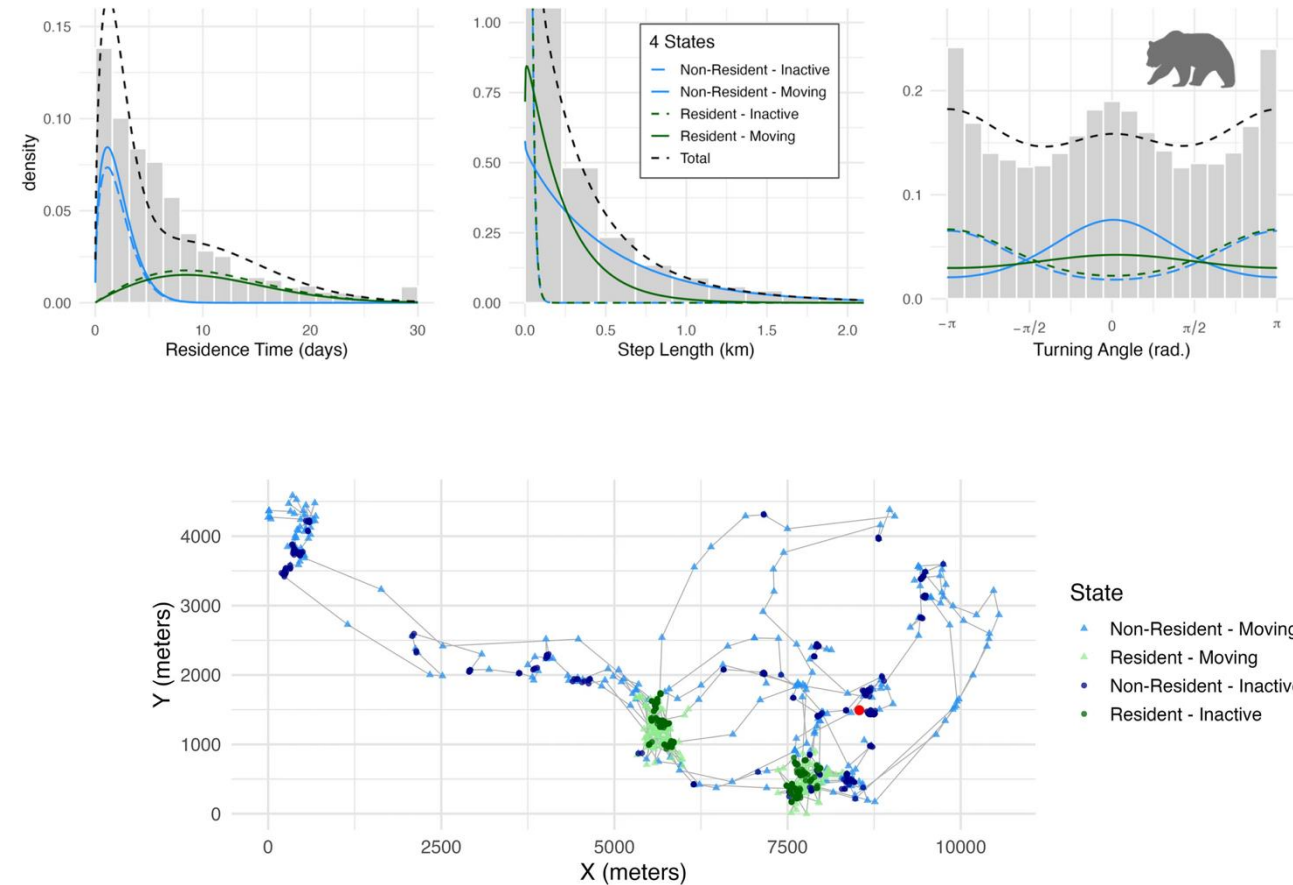
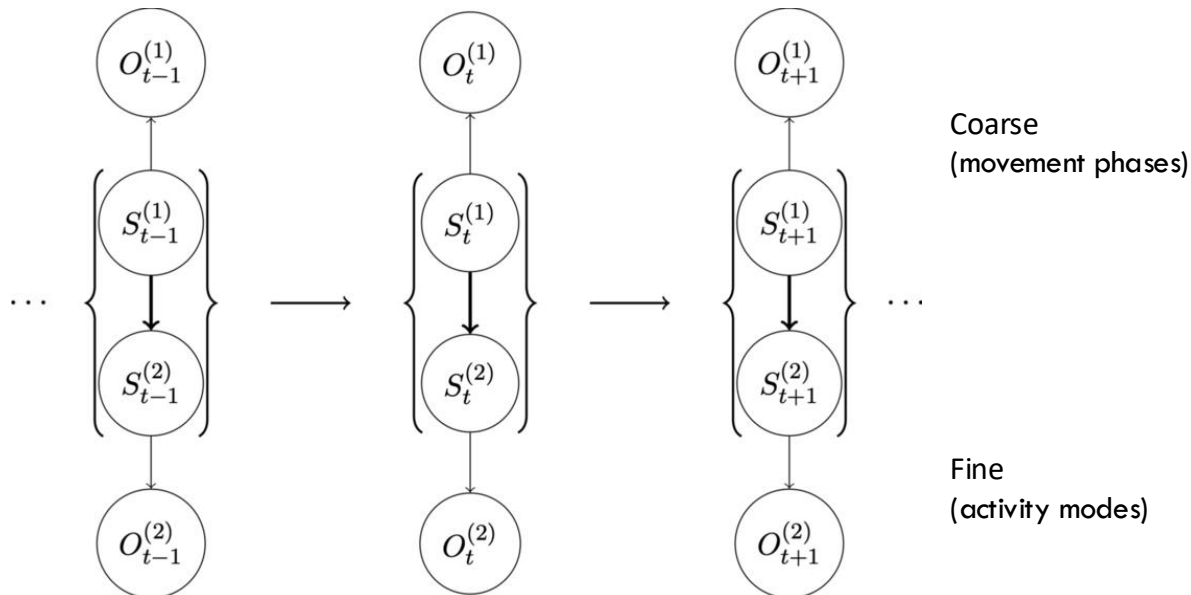
Including multi-scale data

- Data at different time scales using hierarchical HMMs (HHMMs)



Including multi-scale data

- Data at same time scale, but estimating different scales of movement patterns using asymmetric coupled HMMs (ACHMMs)



Time for live-coding



Regularize tracks, estimates states w/ SSM and HMM,
incorporate covariates, apply multiple imputation